https://github.com/mengly-eg/OOP_SE_Y2S2.git

EX1

Material > Week_1 > LAB01 > L01EX1.java > ...

You, 1 hour ago | 1 author (You)

```java
package Material.Week_1.LAB01;

public class L01EX1 {
    public static void main(String[] args) {
        System.out.println(x: "Hello Mengly!");
    }
}
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\eangm\AppData\Roaming\Code\User\workspaceStorage\505b441d42deaadc0eb2a041bfa0fa59\redhat.java\jdt_ws\OOP_c3ea913c\bin' 'Material.Week_1.LAB01.L01EX1'
Hello Mengly!
PS N:\School\Y2-S2\OOP>
```

EX2

Material > Week_1 > LAB01 > L01EX2.java > Java > L01EX2 > main(String[] args)

```java
public class L01EX2 {
    public static void main(String[] args) {
        System.out.println(x: "This is \n break line");
        System.out.println(x: "\t This is Tab");
        System.out.println(x: "This is \'Single Quote\'");
        System.out.println(x: "This is \"Double Quote\"");
        System.out.println(x: "This is \\sign");
        System.out.println(x: "This is \\\\sign");
        // This is a comment        You, 1 hour ago • Done EX4 ...
        /* This is a multi-line comment
           It can span multiple lines
        */
    }
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\eangm\AppData\Roaming\Code\User\workspaceStorage\505b441d42deaadc0eb2a041bfa0fa59\redhat.java\jdt_ws\OOP_c3ea913c\bin' 'Material.Week_1.LAB01.L01EX2'
This is
 break line
         This is Tab
This is 'Single Quote'
This is "Double Quote"
This is \sign
This is \\sign
PS N:\School\Y2-S2\OOP>
```

EX3 Part A

Material > Week_1 > LAB01 > L01EX3.java > Java > L01EX3

You, 1 hour ago | 1 author (You)

```java
package Material.Week_1.LAB01;

public class L01EX3 {
    public static void main(String arg[]) {
        System.err.println("=====Part A=====");
        System.out.println("******************");
        System.out.println(" *****************");
        System.out.println("  ***************");
        System.out.println("   *************");
        System.out.println("    ***********");
        System.out.println("     *********");
        System.out.println("      *******");
        System.out.println("       *****");
        System.out.println("        ***");
        System.out.println("         *");
        System.out.println("\n\n\n");
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\eangm\AppData\Roaming\Code\User\workspaceStorage\505b441d42deaadc0eb2a041bfa0fa59\redhat.java\jdt_ws\OOP_c3ea913c\bin' 'Material.Week_1.LAB01.L01EX3'
=====Part A=====
******************
 *****************
  ***************
   *************
    ***********
     *********
      *******
       *****
        ***
         *
```

EX3 Part B

Material > Week_1 > LAB01 > L01EX3.java > Java > L01EX3

```java
public class L01EX3 {
    public static void main(String arg[]) {
        System.err.println("=====Part B=====");
        System.out.println("********************");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("*                  *");
        System.out.println("********************");
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  n:; cd 'n:\School\Y2-S2\OOP'; & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-ho
=====Part B=====
********************
*                  *
*                  *
*                  *
*                  *
*                  *
*                  *
*                  *
*                  *
*                  *
*                  *
********************
```

EX3 Part C

Material > Week_1 > LAB01 > L01EX3.java > Java > L01EX3

```java
public class L01EX3 {        You, 1 hour ago • Done EX4 …
    public static void main(String arg[]) {
        System.err.println(x: "=====Part C=====");
        System.out.println(x: "1 2 3 4 5");
        System.out.println(x: "2 3 4 5 6");
        System.out.println(x: "3 4 5 6 7");
        System.out.println(x: "4 5 6 7 8");
        System.out.println(x: "5 6 7 8 9");
    }
}
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  n:; cd 'n:\School\Y2-S2\OOP'; & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-ho
*                   *
*                   *
********************




=====Part C=====
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
PS N:\School\Y2-S2\OOP>
```

EX4 Part A

Material > Week_1 > LAB01 > L01EX4.java > Language Support for Java(TM) by Red Hat > L01EX4 > main(String[])

```java
You, 1 hour ago | 1 author (You)
package Material.Week_1.LAB01;

You, 1 hour ago | 1 author (You)
public class L01EX4 {
    Run main | Debug main | Run | Debug
    public static void main(String args[]) {
        //Part A
        System.out.println("=".repeat(count: 23) + "  Part A  " + "=".repeat(count: 23));
        System.out.print(s: "     |");
        for (int i = 1; i <= 10; i++) {
            System.out.printf(format: "%4d|", i);
        }
        System.out.println();
        System.err.println("-".repeat((11*5)+1));


        for (int i = 1; i <= 5; i++) {
            System.out.printf(format: "%4d |", i);
            for (int j = 1; j <= 10; j++) {
                System.out.printf(format: "%4d|", i * j);
            }
            System.out.println();
        }
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
'C:\Users\eangm\AppData\Roaming\Code\User\workspaceStorage\505b441d42deaadc0eb2a041bfa0fa59\redhat.java\jdt_ws\OOP_c3ea913c\bin' 'Material.Week_1.
LAB01.L01EX4'
======================  Part A  =======================
     |   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
-----------------------------------------------------
   1 |   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
   2 |   2|   4|   6|   8|  10|  12|  14|  16|  18|  20|
   3 |   3|   6|   9|  12|  15|  18|  21|  24|  27|  30|
   4 |   4|   8|  12|  16|  20|  24|  28|  32|  36|  40|
   5 |   5|  10|  15|  20|  25|  30|  35|  40|  45|  50|
```

EX4 Part B

Material > Week_1 > LAB01 > L01EX4.java > Language Support for Java(TM) by Red Hat > L01EX4 > main(String[])

```java
public class L01EX4 {
    public static void main(String args[]) {
        System.out.print(s: "\n\n\n");
        You, 1 hour ago • Done EX4
        //Part B
        System.out.println("=".repeat(count: 23) + "  Part B  " + "=".repeat(count: 23));
        String format = "| %-15s | %-8s | %-20s | %-4s |%n";
        String line = "+-----------------+----------+----------------------+------+%n";

        System.out.printf(line);
        System.out.printf(format, ...args: "Country", "Name", "Profession", "Age");
        System.out.printf(line);

        System.out.printf(format, ...args: "Germany", "Michael", "Computer Engineer", "19");
        System.out.printf(line);
        System.out.printf(format, ...args: "England", "Robert", "Artist", "34");
        System.out.printf(line);
        System.out.printf(format, ...args: "United Kingdom", "Julia", "Designer", "42");
        System.out.printf(line);
        System.out.printf(format, ...args: "United States", "Jo", "Actor", "21");
        System.out.printf(line);
    }
}
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
'C:\Users\eangm\AppData\Roaming\Code\User\workspaceStorage\505b441d42deaadc0eb2a041bfa0fa59\redhat.java\jdt_ws\OOP_c3ea913c\bin' 'Material.Week_1.
LAB01.L01EX4'
======================  Part B  ======================
+-----------------+----------+----------------------+------+
| Country         | Name     | Profession           | Age  |
+-----------------+----------+----------------------+------+
| Germany         | Michael  | Computer Engineer    | 19   |
+-----------------+----------+----------------------+------+
| England         | Robert   | Artist               | 34   |
+-----------------+----------+----------------------+------+
| United Kingdom  | Julia    | Designer             | 42   |
+-----------------+----------+----------------------+------+
| United States   | Jo       | Actor                | 21   |
+-----------------+----------+----------------------+------+
PS N:\School\Y2-S2\OOP>
```

EX5 Part A, B, C

Material > Week_1 > LAB01 > L01EX5.java > ...

You, 27 minutes ago | 1 author (You)

```java
package Material.Week_1.LAB01;

You, 27 minutes ago | 1 author (You)
public class L01EX5 {
    Run main | Debug main | Run | Debug
    public static void main(String args[]) {
        //Part A
        System.out.println("=".repeat(count: 15) + "  Part A  " + "=".repeat(count: 15));
        String textA = "I Love My Home Country";
        System.out.println("text length is : " + textA.length() + "\n\n");


        //Part B
        System.out.println("=".repeat(count: 15) + "  Part B  " + "=".repeat(count: 15));
        String textB = "I Love My Little Country";
        System.out.println("Result: " + textB.substring(beginIndex: 7) + "\n\n");


        //Part C
        System.out.println("=".repeat(count: 15) + "  Part C  " + "=".repeat(count: 15));
        String textC = "Hi Students!";
        System.out.println("'i' is at index: " + textC.indexOf(ch: 'i'));
        System.out.println("'u' is at index: " + textC.indexOf(ch: 'u'));
        System.out.println("'e' is at index: " + textC.indexOf(ch: 'e'));
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
===============  Part A  ===============
text length is : 22


===============  Part B  ===============
Result: My Little Country


===============  Part C  ===============
'i' is at index: 1
'u' is at index: 5
```

EX 5 Part D, E

Material > Week_1 > LAB01 > L01EX5.java > Language Support for Java(TM) by Red Hat > L01EX5 > main(String[])

```java
public class L01EX5 {
    public static void main(String args[]) {
        //Part D
        System.out.println("\n\n" + "=".repeat(count: 15) + "  Part D  " + "=".repeat(count: 15));
        String textD1 = "Hi Students!";
        String textD2 = "Teacher";
        String textD3 = "Student";
        if (textD1.equals(textD2)) {
            System.out.println("Text1 is equal to Text2: " + true);
        } else {
            System.out.println("Text1 is equal to Text2: " + false);
        }

        if (textD1.equals(textD3)) {
            System.out.println("Text1 is equal to Text3: " + true);
        } else {
            System.out.println("Text1 is equal to Text3: " + false);
        }

        //Part E
        System.out.println("\n" + "=".repeat(count: 15) + "  Part E  " + "=".repeat(count: 15));
        if (textD1.contains(textD2)) {
            System.out.println("Text1 contains Text2: " + true);
        } else {
            System.out.println("Text1 contains Text2: " + false);
        }
        You, 30 minutes ago • Update EX5 …
        if (textD1.contains(textD3)) {
            System.out.println("Text1 contains Text3: " + true);
        } else {
            System.out.println("Text1 contains Text3: " + false);
        }
    }
}
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\OOP>  n:; cd `n:\School\Y2-S2\OOP`; & `C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe` `-XX:+ShowCodeDetailsInExceptionMessage
===============  Part D  ===============
Text1 is equal to Text2: false
Text1 is equal to Text3: false

===============  Part E  ===============
Text1 contains Text2: false
Text1 contains Text3: true
PS N:\School\Y2-S2\OOP>
```

EX 6

Material > Week_1 > LAB01 > L01EX6.java > ...

```java
You, 19 minutes ago | 1 author (You)
package Material.Week_1.LAB01;

You, 19 minutes ago | 1 author (You)
class Employee {
    String name, profession, gender;
    int age, salary;
}
You, 19 minutes ago | 1 author (You)
class Car {
    String brand, model;
    int year, price, mileage;
}
You, 19 minutes ago | 1 author (You)
class BankAccount {
    String accountName, accountNumber;
    double deposit, withdrawal, balance;
}
You, 19 minutes ago | 1 author (You)
class Video {
    String title, director;
    int duration, releaseYear, rating;
}
You, 19 minutes ago | 1 author (You)
public class L01EX6 {
    Run main | Debug main | Run | Debug
    public static void main(String args[]) {

        Employee employee1 = new Employee();
        Car car1 = new Car();

        employee1.name = "Mengly";
        employee1.profession = "Software Engineer";
        employee1.gender = "Male";
        employee1.age = 30;
        employee1.salary = 80000;

        car1.brand = "Mercedes-Benz";
        car1.model = "SLK 55";
        car1.year = 2018;
        car1.price = 55000;
        car1.mileage = 15000;
    }
}
```